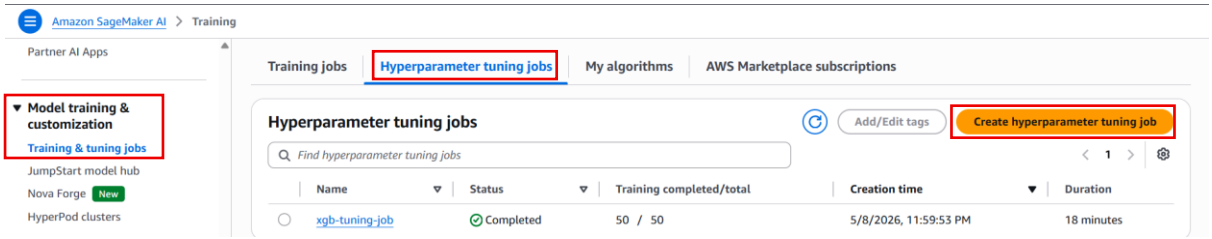
Hyperparameter Tuning in the UI

To Begin with the Lab

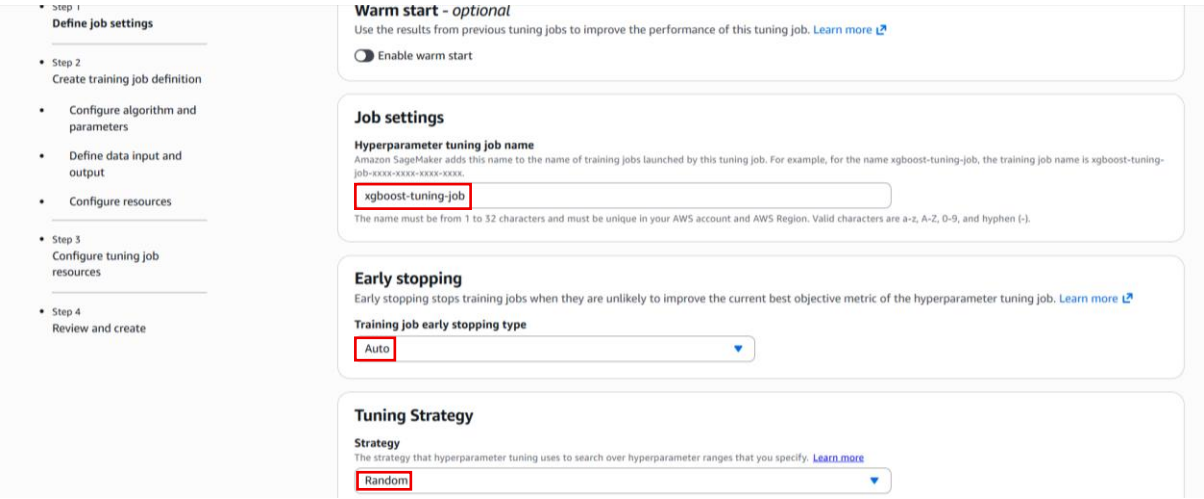
Summary of the Lab

This lab walks through creating a hyperparameter tuning job in Amazon SageMaker using the built-in XGBoost algorithm. You configure a tuning job with Bayesian optimization and automatic early stopping, define training data stored in Amazon S3, and set compute resources using an ml.m4.xlarge instance. The lab also demonstrates configuring tuning limits, including parallel and maximum training jobs, to efficiently optimize model performance through automated experimentation.

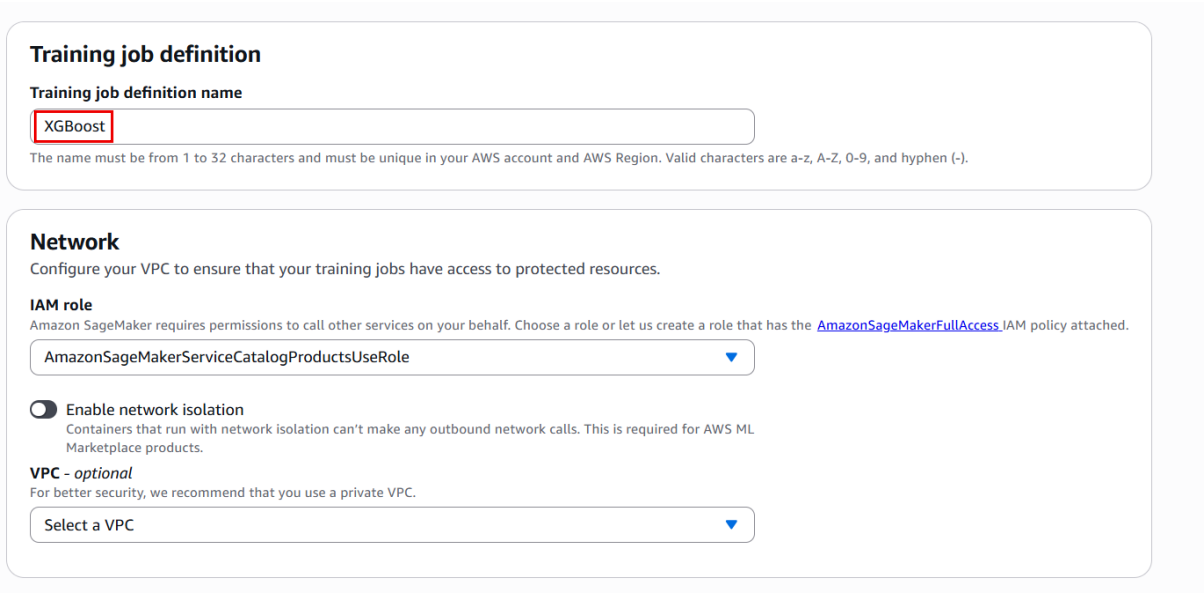
- Go to Model training & customization
- Select Training & tuning jobs
- Click Hyperparameter tuning jobs
- Click Create hyperparameter tuning job



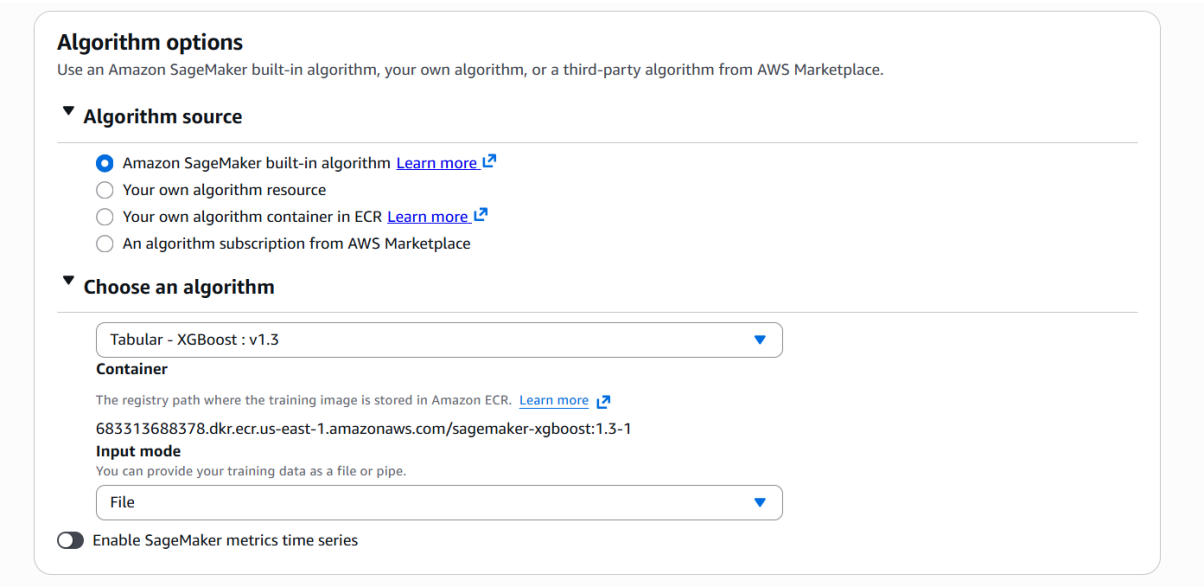
- Configure the following:
- Warm Start (Optional)
- Job Name
    - xgboost-tuning-job
- Early Stopping
    - Set to Auto
- Choose Tuning Strategy
    - Choose Bayesian



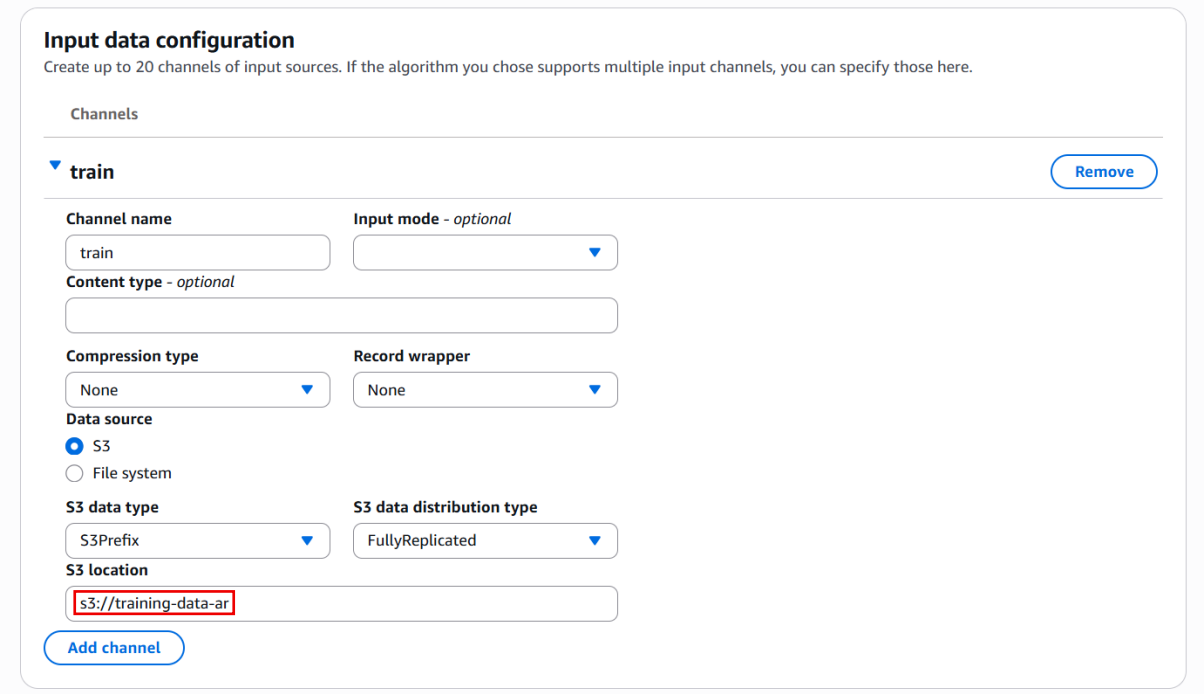
- Then Click Next
- Configure Training Job Definition
    - Click Add Training Job Definition
- Training Job Name
    - XGBoost



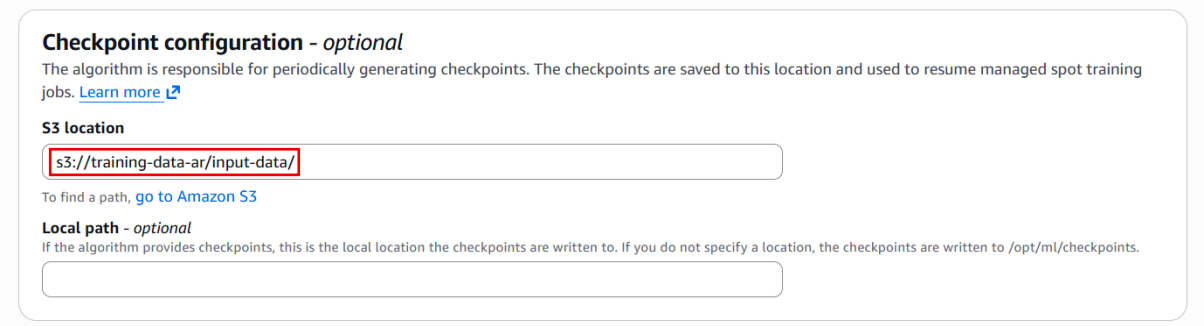
- Algorithm
- Select:
    - Amazon SageMaker built-in algorithm
- Choose:
    - XGBoost
- Input Mode
- Select:
    - File



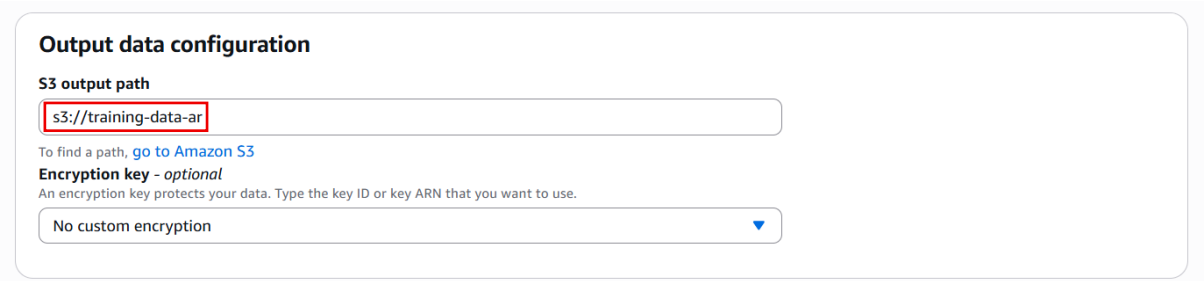
- Configure Data Channels
- Training Data Location
- Input Data Configuration
    - s3://your-bucket



- Check point configuration

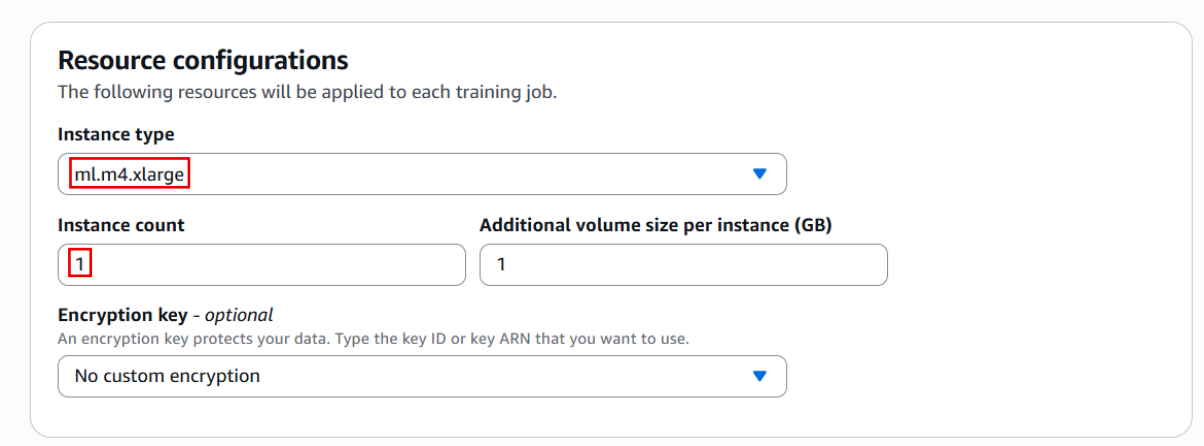


- Output data Configuration

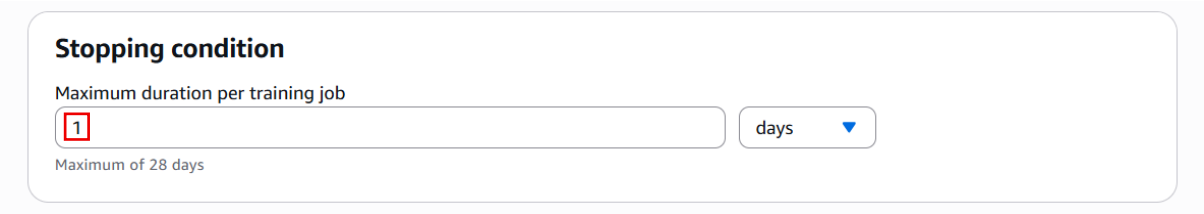


- Configure Compute Resources
- Instance Type
    - ml.m4.xlarge

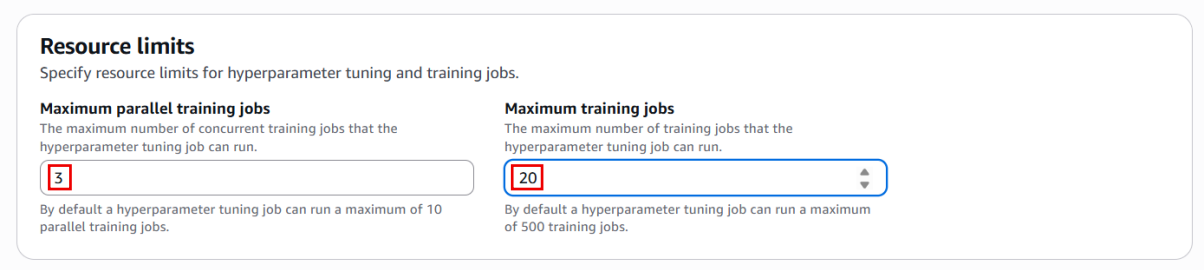
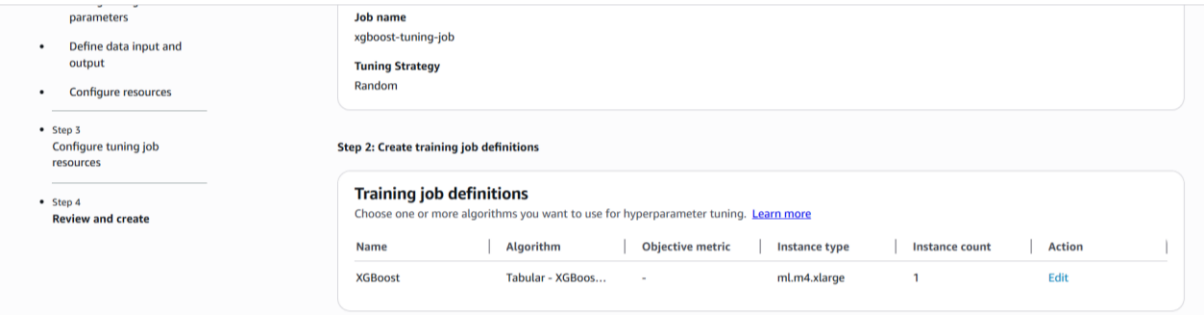
- Instance Count
- Set to:
    - 1



- Stopping Condition



- Configure Tuning Limits
- Example settings:
- Setting Value
- Parallel Jobs-3
- Maximum Training Jobs-20



- Here you can review and create your tuning